

NELSON, BC, Canada

WMO: 717760

Lat: 49.491N

Lon: 117.305W

Elev: 535

StdP: 95.06

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-11.5	-8.9	-16.2	1.0	-9.7	-13.3	1.3	-7.4	7.4	-2.1	6.5	-1.4	3.1	250	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	32.3	18.6	30.2	18.0	28.2	17.5	20.1	28.3	19.1	27.3	18.2	26.1	3.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	13.4	22.8	16.4	12.4	21.5	15.4	11.7	20.6	60.0	28.5	56.5	27.4	53.5	26.2	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.9	5.2	DB	-13.8	35.5	3.1	1.6	-16.0	36.6	-17.8	37.6	-19.5	38.5	-21.7	39.6	(4)
(5)			WB	-14.3	21.8	3.0	1.2	-16.5	22.7	-18.2	23.4	-19.8	24.1	-22.0	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-0.7	0.5	4.2	8.4	12.9	16.0	20.5	20.2	15.1	8.2	3.0	-1.0	(6)
(7)		DBStd	8.20	3.62	3.32	2.97	2.76	3.24	3.05	3.06	2.91	3.16	3.02	3.12	3.56	(7)
(8)		HDD10.0	1480	333	265	181	64	11	1	0	0	2	71	210	342	(8)
(9)		HDD18.3	3591	591	498	439	299	170	82	15	16	106	313	460	601	(9)
(10)		CDD10.0	1107	0	0	1	15	101	182	324	315	154	16	0	0	(10)
(11)		CDD18.3	176	0	0	0	0	2	13	81	72	8	0	0	0	(11)
(12)		CDH23.3	2099	0	0	0	2	69	212	926	781	109	0	0	0	(12)
(13)		CDH26.7	764	0	0	0	0	11	60	382	288	23	0	0	0	(13)
(14)	Wind	WSAvg	2.3	2.6	2.4	2.4	2.3	2.2	2.3	2.2	2.1	2.0	2.2	2.3	2.4	(14)
(15)	Precipitation	PrecAvg	724	90	51	55	48	62	69	37	33	42	65	81	93	(15)
(16)		PrecMax	917	145	104	111	80	129	130	86	88	133	101	158	153	(16)
(17)		PrecMin	579	24	9	14	20	16	37	1	6	3	5	27	47	(17)
(18)		PrecStd	109	25	24	27	19	28	20	23	25	30	25	37	28	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	9.2	16.2	22.2	28.2	31.4	35.3	34.1	29.3	19.7	10.9	6.2	(19)
(20)			MCWB	3.8	5.5	8.8	12.2	15.9	18.1	19.1	18.7	17.6	13.2	8.7	5.1	(20)
(21)		2%	DB	4.3	6.9	12.8	19.0	25.2	28.2	32.9	31.9	26.3	16.9	9.4	4.5	(21)
(22)			MCWB	3.2	4.1	7.1	10.9	14.6	17.0	18.7	18.6	16.4	11.8	8.1	3.6	(22)
(23)		5%	DB	3.6	5.4	10.7	16.8	22.7	25.7	30.8	29.9	24.0	14.8	8.1	3.6	(23)
(24)			MCWB	2.7	3.5	6.0	9.6	13.5	15.9	18.4	17.9	15.6	10.8	6.9	2.8	(24)
(25)		10%	DB	2.9	4.2	8.7	14.5	20.3	23.1	28.6	27.9	21.6	13.0	7.1	2.8	(25)
(26)			MCWB	2.2	2.9	5.2	8.4	12.3	15.1	17.9	17.4	14.5	9.9	6.1	2.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.5	6.3	9.4	13.2	16.7	19.6	21.8	21.5	18.8	14.7	9.8	5.5	(27)
(28)			MCDWB	5.0	7.9	14.4	20.5	25.1	28.6	30.4	29.6	25.9	17.8	10.2	6.0	(28)
(29)		2%	WB	3.6	4.8	7.8	11.5	15.2	17.9	20.4	19.9	17.4	13.2	8.3	3.9	(29)
(30)			MCDWB	4.0	6.0	11.5	17.4	22.8	25.5	28.5	28.2	24.5	15.8	8.9	4.3	(30)
(31)		5%	WB	2.9	3.9	6.7	10.2	14.2	16.8	19.4	19.0	16.3	11.6	7.1	3.0	(31)
(32)			MCDWB	3.4	5.0	9.7	15.5	21.1	23.6	27.7	26.9	22.4	13.9	7.9	3.4	(32)
(33)		10%	WB	2.3	3.1	5.6	9.0	13.1	15.7	18.5	18.1	15.3	10.3	6.1	2.3	(33)
(34)			MCDWB	2.8	4.0	8.1	13.3	19.0	21.6	26.6	26.0	20.7	12.4	7.0	2.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.8	6.3	8.3	11.0	12.6	12.1	14.8	14.6	12.2	8.3	4.6	3.5	(35)
(36)			MCDBR	4.0	7.7	12.1	15.7	17.8	17.4	18.8	18.1	16.1	11.3	5.8	3.9	(36)
(37)			MCWBR	3.1	5.2	7.0	8.1	8.0	7.4	7.2	7.4	7.9	6.7	4.2	3.2	(37)
(38)		5% WB	MCDWR	3.7	6.6	10.7	14.0	15.4	15.2	16.4	15.9	14.8	9.7	4.7	3.5	(38)
(39)			MCWBR	3.0	4.7	6.5	7.7	7.3	7.0	7.0	7.3	7.7	6.1	3.6	3.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.279	0.284	0.300	0.357	0.384	0.379	0.371	0.386	0.348	0.323	0.298	0.272	(40)	
(41)		taud	2.276	2.283	2.293	2.251	2.316	2.375	2.411	2.371	2.471	2.517	2.432	2.298	(41)	
(42)		Ebn at Noon	768	864	912	886	875	880	882	850	852	817	750	722	(42)	
(43)		Edh at Noon	71	90	106	124	122	116	110	110	90	72	62	60	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.94	1.86	2.73	4.25	5.18	5.45	6.57	5.51	3.85	2.20	0.99	0.74	(44)	
(45)		RadStd	0.13	0.23	0.41	0.40	0.45	0.51	0.54	0.46	0.43	0.31	0.15	0.10	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	+1.44	N/A	N/A	N/A	N/A	+113	+43	(46)
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+65	N/A	(47)

Nomenclature: See separate page